Manish Neupane

Brooklyn, SD | (605) 691-2858 | manishneupane0909@gmail.com | GitHub | www.mneupane.com

Professional Summary

Software engineer and dual-major Physics & Computer Science graduate (May 2026) with 2+ years of experience building automation, full-stack applications, and data-driven systems in industry and research environments. Experienced with Python, JavaScript, React, FastAPI, Django REST Framework, SQL, MongoDB, Docker, and AWS.

Education

South Dakota State University (SDSU), Brookings, SD — Aug 2021 – May 2026

B.S. in Physics & Computer Science

- Earned URSCAD Outstanding Student Achievement in Research (Spring 2022, Spring 2024) and Outstanding First Year Residential Assistant recognition.
- Completed coursework in Data Structures & Algorithms, Operating Systems, Machine Learning, Software Engineering, and Database Systems.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, C#, Java, Go, C/C++

Frameworks: React, FastAPI, Django REST Framework, .NET

Databases/Tools: PostgreSQL, MongoDB, Redis, Docker, AWS, Git, REST APIs, JWT, Celery

Concepts: Data Structures & Algorithms, OOP, TDD, Clean Code, Hardware-Software Integration

Experience

Daktronics, Product Engineering Intern, Brookings, SD — May 2024 – Present

- Optimized production test procedures and supported NPI workflows, increasing throughput by 20%.
- Debugged LED display control logic and resolved failure patterns, reducing hardware rejection rates by 10%.
- Analyzed defect trends across display and control systems to improve reliability and support product improvements.

Starship Technologies, Software & Operations Technician, Brookings, SD — Jan 2022 – May 2023

- Developed and deployed Python and Go scripts for firmware updates and runtime code modifications on autonomous delivery robots.
- Diagnosed failures across sensors, GPS, motors, cameras, and batteries to maintain 98% uptime for a 10-unit fleet.
- Analyzed fleet telemetry data to identify inefficiencies and improve delivery efficiency and user satisfaction by 15%.

South Dakota State University, Undergraduate Research Assistant, Brookings, SD — Aug 2021 – Present

- Architected an automated Python pipeline using SciPy and VSM integration to batch-process 500+ magnetometer datasets.
- Reduced analysis time from 2 days to 4 hours, improving workflow efficiency by 80%.
- Co-authored 3 publications/abstracts and presented findings at APS March Meetings in 2023 and 2024.

Projects

Secure Path Optimization Platform — Django REST Framework, React, PostgreSQL, scikit-learn, Celery, Redis

- Built a full-stack fraud detection platform that integrated Plaid API data ingestion and a scikit-learn classifier to score transaction risk.
- Implemented asynchronous background processing with Celery and Redis and secured endpoints with token-based authentication.

ElementX — React, FastAPI, Python, MongoDB, SciPy, Docker

- Built a full-stack materials science platform for stoichiometry calculations, XRD peak detection, and magnetic property extraction.
- Engineered an async FastAPI backend with MongoDB, JWT authentication, and Docker Compose deployment.

Physics-Informed Neural Networks (PINNs) — Python, PyTorch, NumPy

- Developed a neural PDE solver that achieved a 25% speedup over classical finite-difference methods and reduced simulation compute time by 40%.